Universal Design and Multimethod Approaches to Item Review

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Test items undergo multiple iterations of review before states and vendors deem them acceptable to be placed in a live statewide assessment. This article reviews three approaches that can add validity evidence to states' item review processes. The first process is a structured sensitivity review process that focuses on universal design considerations for items. The second method is a series of statistical analyses intended to increase the limited amount of information that can be derived from analyses on low-incidence populations (such as students who are blind, deaf, or have cognitive disabilities). Finally, think aloud methods are described as a method for understanding why particular items might be problematic for students.

Keywords: assessment, disability, large-scale assessment

Recent changes in Federal legislation (including the No Child Left Behind Act of 2001) have placed greater emphasis on accountability via large-scale tests. Previously exempted students now are included and have scores reported in statewide large-scale testing systems. Schools, states, and districts are accountable for the educational success of all of their students. Although large-scale tests are only one of several indicators for which states must demonstrate "Adequate Yearly Progress," such tests have been the objects of attention in both academic research (Linn, Baker, & Betebenner, 2002) and popular press (Henriques, 2003).

In the current era of high-stakes testing, some stakeholders have expressed concern that large-scale assessments do not provide an accurate measure of students' knowledge and skills (see Fuhrman & Elmore, 2004). Therefore, states and test companies have begun to search for solutions to validity questions when tests are administered to diverse populations. One proposed solution is the use of Universal Design of Assessment elements. Universal Design of Assessment (UDA) is broadly defined as assessments that are "designed and developed from the beginning to allow participation of the widest possible range of students, and to result in valid inferences about performance for all students who participate in the assessment" (Thompson, Johnstone, & Thurlow, 2002, p. 5).

The term universal design (UD) was originally used as an architectural concept (Center for Universal Design, n.d.). Philosophies of access and barrier removal were the foundations for early universal design approaches. As part of this design philosophy, ramps, elevators, expanded doorways, signs, bathrooms, and other features do not have to be added or modified at additional expense after the completion of a building. Rather, as part of UD, these features are sketched into structures' blueprints from the beginning. The promise of UD is that some of the same architectural features that accommodate people with disabilities also benefit many others, including senior citizens, families with young children, and delivery people.

The term UD has emerged as a cross-disciplinary concept and has a growing literature base with successful interventions. Research in architecture (Center for Universal Design, n.d.; Iwarsson & Stahl, 2003), instruction (Acrey, Johnstone, & Milligan, 2005; Belch, 2005; Field, Sarver & Shaw, 2003; Renzagalia, Karvonen, Drasgow, & Stoxen, 2003; Smith & Meyen, 2003), and assessment (Dolan, Hall, Banerjee, Chun, & Strangman, 2005; Ketterlin-Geller, 2005) has demonstrated the potential of UD approaches to increase access to environments, learning, and assessment for persons with disabilities. Approaching tests from a universal design framework means that tests: have inclusive assessment populations, measure what they intend to measure, contain minimal bias, have clear and understandable instructions and procedures, are amenable to accommodations, are comprehensible, and are legible (Thompson et al., 2002). Universal design as a philosophy is broad based, and therefore may include a variety of

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strategies to help understand how item design affects student performance.

Review of Literature

The overriding aim of universal design, and seemingly all attempts to improve assessment, is to improve validity. The current era of high stakes testing is rife with complexity. Inclusive testing environments are challenging because issues such as student disability or English language learner status may cloud a student's ability to demonstrate her or his knowledge on tests. Thus, tests that are not designed with inclusiveness in mind may not adequately discriminate between students who understand constructs but are affected by design features from students who simply do not understand constructs. In this vein, universal design is one way to improve the validity of tests for all students by considering "best practices" in comprehension, reading, vision, measurement, and assessment of specific populations. For example, ideas about fidelity to intended constructs come from best practice in assessment. Similarly, approaches for legible print are derived from literature on vision and reading print (see Thompson et al., 2002 for a complete discussion on UDA's multidisciplinary roots).

Many of the approaches to universal design of assessments are not new. In fact, many were derived directly from the American Educational Research Association (AERA), American Psychological Association (APA), and National Council on Measurement in Education (NCME) (1999) *Standards for Educational and Psychological Testing.* For example, the *Standards* address inclusiveness by requiring that "all examinees be given a comparable opportunity to demonstrate their standing on the construct(s) the test is intended to measure" (p. 74). Also, universal design approaches encourage test designers to consider *all* examinees when designing items and tests.

The AERA, APA, and NCME standards also address validity by challenging test designers to minimize construct-irrelevant variance. Construct-irrelevant variance was described by Messick (1989, 1996) as the effects on scores produced by activities or processes that are extraneous to the intended constructs. Haladyna and Downing (2004) noted that there are two main sources of construct-irrelevant variance. First, systematic error may occur for all members of a particular group, resulting in underrepresentation of student achievement for an entire group. Abedi, Leon, and Mirocha (2001), for example, noted how features of large-scale assessments may underestimate the achievement of English language learners and Grise, Beattie, and Algozzine (1982) found that design features may underestimate the achievement of students with disabilities.

Ensuring validity of items is not an easy process. Downing and Haladyna (1997) found that 11 types of evidence must be present before items may be considered valid, including: a definition of content, test specifications, item writer training, adherence to item-writing principles, classification of cognitive behaviors tested, item content verification, item editing, sensitivity reviews, item tryouts, key validation, and a test security plan. Among the types of evidence described by Downing and Haladyna, expert/bias reviews is one method that has been addressed significantly in literature. Such reviews allow experts to examine items for quality and sensitivity prior to placement in an operational test. Zieky (2006) described several steps for conducting item reviews on licensure tests. Many of his suggestions are also important in reviewing items for K-12 large-scale assessment, including:

- Establishing an advisory group specifically for fairness issues.
- Including only content that is clearly justifiable in the test.
- Providing guidelines to reviewers to avoid sexist, racist, stereotyped, or offensive language, illustrations, or stimulus materials. Train item writers using the same guidelines.
- Ensuring that linguistic demands are concordant with the test's purpose.
- Treating test takers equally regardless of personal characteristics that are not relevant to the test.
- Providing accommodations for people with disabilities (or others as allowable).

According to Zieky (2006), there is no perfect approach for ensuring that test items are fair to test takers. Nonetheless, a structured and documented approach to reviewing items may increase the quality and fairness for the assessment population; doing so will help ensure that assessments reflect real, not construct-irrelevant differences in populations. Fairness reviews, along with careful editing of items for content (Webb, 2006) and grammar (provided ample time is given for revisions) may improve items in their early stages, when it is easiest to fix questionable items (Baranowski, 2006).

Once items have been prescreened and do enter pilot stages, item bias may be reduced by conducting analyses of test results. The AERA, APA, and NCME (1999) *Standards* encourage stakeholders to

> research whether any items are more difficult for students from particular subpopulations. This can be accomplished through the administration of a field test that can help determine item difficulty and ability to discriminate among test takers of different standing on the scale. (p. 39)

The purpose of statistical analyses is to provide states and vendors with an empirical check on the fairness of items (Zieky, 2006). In this case, items that may be questionable are those that behave differently within a particular test for particular populations.

Marion's (2006) "life of an ELA item" (see Figure 1) demonstrates that an item is subjected to multiple review processes before finding its way into an operational test. This article reviews three approaches that can be applied to item review processes, potentially at multiple points. Two of the approaches recommended are simply different ways of approaching existing practices of item review. The third approach (think aloud methods) is a data collection method that may help answer additional questions about item design. Ericsson and Simon (1995) noted that think aloud verbalizations are an important data source for understanding cognitive activities. Such approaches can be applied to thinking processes of students on test items (Leighton, 2004). In think aloud activities, verbalizations that take place concurrently with cognitive processes, however, are largely independent of interpretation on the part of the subject (Van Someren, Barnard, & Sandberg, 1994). Researchers often ask follow-up questions of research participants to clarify utterances.

All methods described above come from an explicit grounding in federal

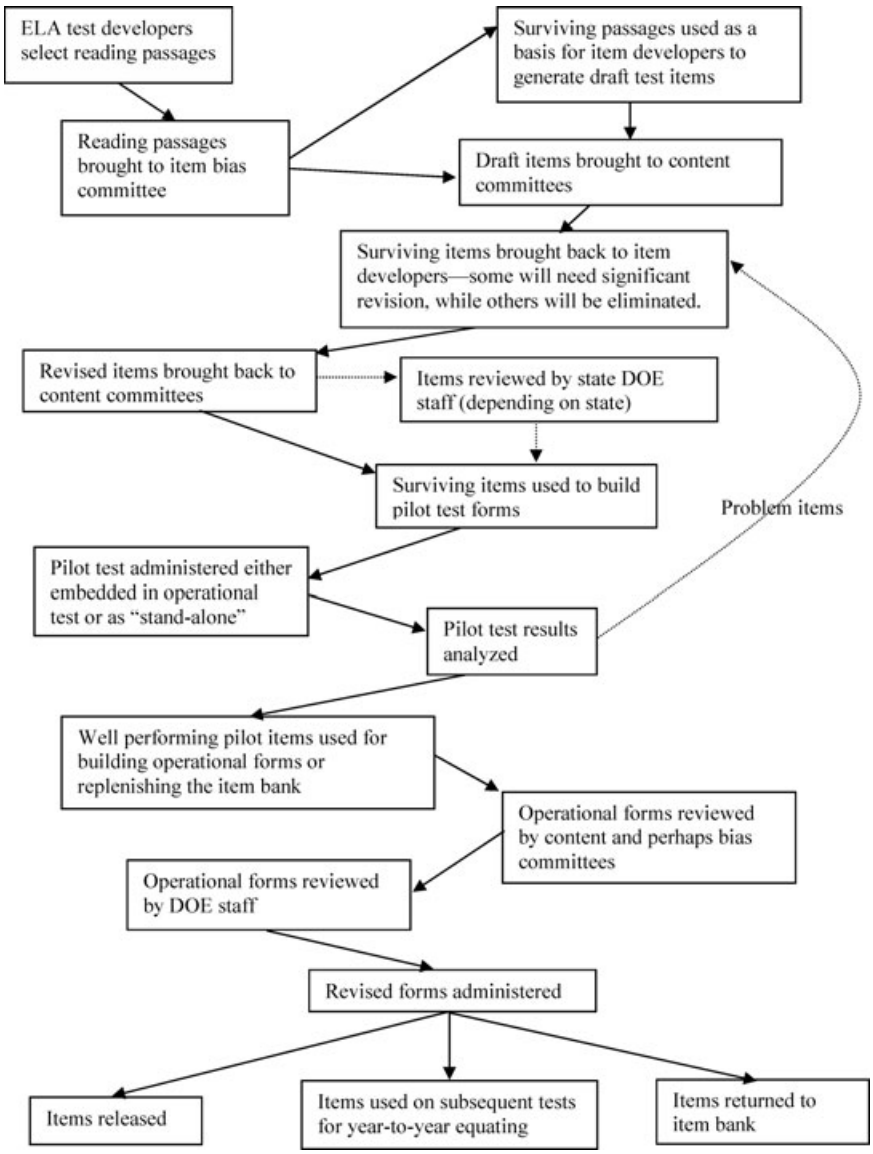


FIGURE 1. The life of an ELA item (reprinted with permission).

policy, which requires the inclusion of diverse students in large-scale assessments and adherence to maintaining the validity of the construct of the content-based assessments. All processes fall under the heading of "universal design" because they seek to improve the validity of assessments for *all* examinees, including subpopulations like students with disabilities.

Overview of UDA Approaches

Three item review processes are reviewed in this article. The methods and results were conducted separately, on separate data sets, but are presented here in the order of the chronology of item review processes. Readers should note that the three approaches have never been used within a single item review session, but are presented this way here as a heuristic for possible inclusion in states' item review processes. The first process reviewed in this article is a structured sensitivity review process that focuses on universal design considerations for items. As seen in Marion's "life of an ELA item" diagram (Figure 1), sensitivity and content reviews often occur soon after items are written. States and vendors typically conduct thorough reviews of items to ensure that they align with content standards and are not potentially biased against particular populations. This article describes an approach to item review that incorporates considerations for "universal design," i.e., the review processes described provide reviewers with a structured process for examining items based on research-based considerations of accessible item design.

The second method is intended to support current analyses that take place after field or operational testing. We propose a series of statistical analyses intended to increase the limited amount of information that can be derived from analyses on low-incidence populations (such as students who are classified as blind or deaf, or who have cognitive impairments). Our approaches are proposed in full recognition that there is no valid way to statistically examine items for low-incidence populations, but these populations are still part of assessment populations and therefore merit consideration. These methods are meant to supplement, not replace, existing analyses conducted by states, and are described in detail below.

Finally, think aloud methods are described as a method for understanding why particular items might be problematic for students. Returning to Marion's (2006) "life of an ELA item," think aloud methods may be introduced early in the item development process (to supplement item review data) or after field-testing has occurred (to understand why particular item statistics may have emerged). The number of items that can be examined using think aloud methods is limited by the time and expense required for the method, but the information derived may help states and vendors to better understand how item design affects how students approach (and solve) items.

The three methods are presented below in their entirety (reporting samples, instruments, analyses, and results). Although this reporting style is not in accordance with APA style, the authors, editors, and reviewers of this article determined it to be an appropriate style for this report. For the purposes of clarity, synopses of entire expert review, statistical analyses, and think aloud methods are presented below.

Expert Review

Shortly after an item is written, it will undergo scrutiny from content and sensitivity reviewers. A preliminary process of item review employs the expertise of community and educational authorities. Such reviews are commonplace in high-stakes testing. Expert reviews aim to minimize bias (Geisinger & Carlson, 1992; Popham, 2001; Popham & Lindheim, 1980) in tests and promote

Considerations for Universally Designed Assessment Items Subject _____ Grade _____ Test form ___ Item #s on this page _1-5___
Initials _____

Star (*) areas of strength and Check (√) areas of concern for each item	Pass -age	Item #1	Item #2	Item #3	Item #4	Item #5	Describe Concerns and Suggestions for items and reading passages	Recommend review by expert or student in Content Area, Specific Disability, Language, Culture	
Item measures its intended construct									
Item respects the diversity of the assessment population • Sensitive to test taker characteristics and experiences (gender, age, ethnicity, SES, region, disability, language) • Avoids content that might unfairly advantage or disadvantage any student subgroup • Other								Expert review?	Student review?
Item has concise and readable text • Commonly used words (except vocabulary tested) • Vocabulary appropriate for grade level • Minimum use of unnecessary words • Technical terms and abbreviations avoided unless tested • Sentence complexity appropriate for grade level • Question to be answered identifiable • Other								Expert review?	Student review?
Item has a clear format for text • Standard typeface • Twelve (12) point minimum size for all print, • High contrast between text and background • Sufficient blank space • Staggered right margins • Other								Expert review?	Student review?
Item has clear visuals (use NA for none) • Visuals are needed to answer the question • Visuals have clearly defined features • High contrast between visuals and background • Visuals are clearly labeled • Other								Expert review?	Student review?
Item allows changes to format without changing meaning or difficulty (check allowed accommodations) • Braille or other tactile format • Sign language interpretation • Oral presentation • Assistive technology • Translation into another language • Other								Expert review?	Student review?
Describe other considerations specific to item on back									

FIGURE 2. Considerations for universally designed assessments.

alignment of curriculum standards with test items (Webb, 1999). Expert reviews, however, may lack structure and often involve subjective definitions of bias. Additionally, reviewers may conduct reviews without specific knowledge of what good and questionable items look like.

Finally, sensitivity review panels may examine the content of items for bias, but may not focus on item design for accessibility. To remedy issues of item reviews, and to give students a "just" opportunity to demonstrate knowledge (AERA, APA, NCME, 1999, p. 74), Thompson, Johnstone, Anderson and Miller (2005) produced a list of considerations that outlines design elements that are necessary for the creation of universally designed large-scale test items. This list of considerations is an extension of Thompson et al.'s (2002) *Elements of Universal Design*. Thompson et al. (2002) reviewed research in a variety of fields (vision, graphic arts, measurement, literacy, special education, and measurement) to produce their original *Elements*. In 2005, Thompson and her colleagues created the *Considerations* document (see Figure 2) based on results from a Delphi study conducted with content area, English language learner (ELL),

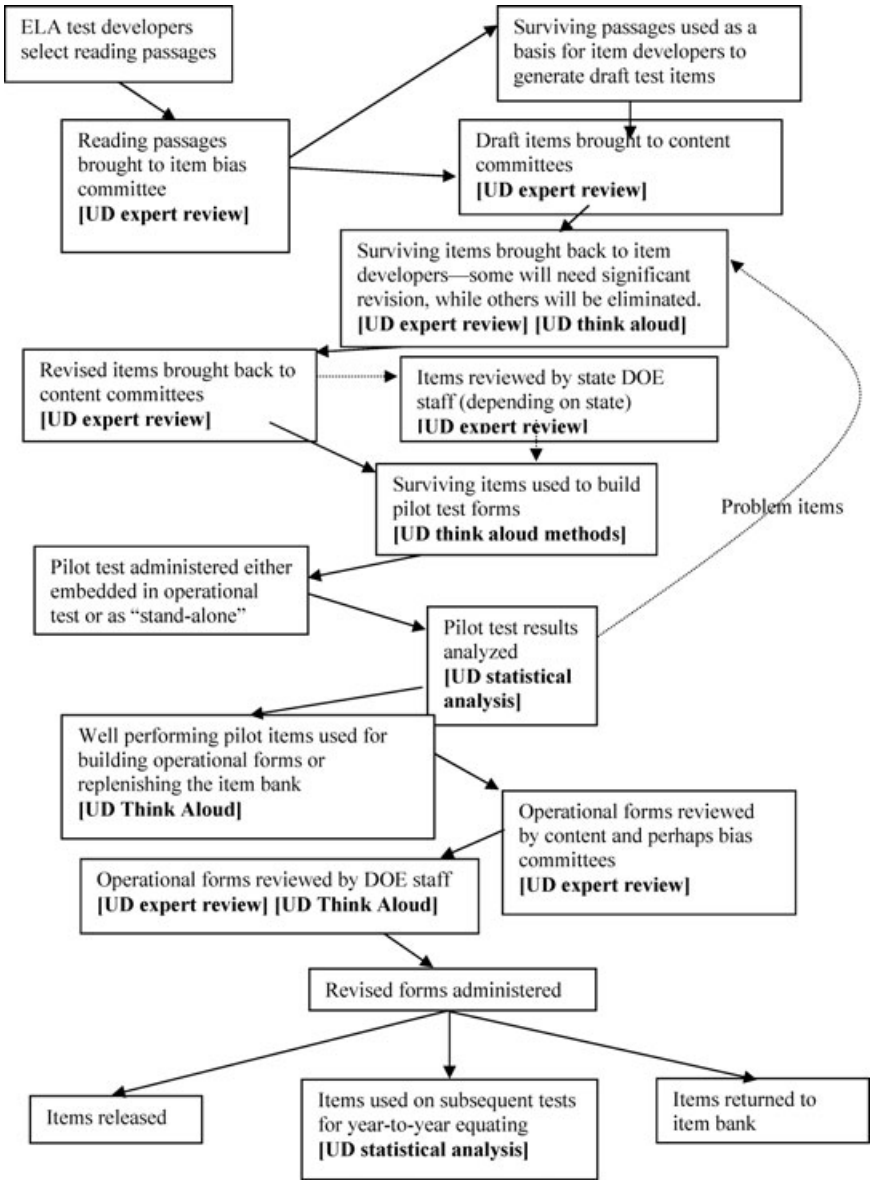


FIGURE 3. "Life of an ELA Item" with potential universal design considerations.

measurement, and disability experts. The intent of the Delphi review was to invite experts to give input on the considerations and modify them accordingly (Alder & Ziglio, 1996). Such considerations are similar to the Center for Applied Special Technology's (CAST) "Bobby" programs that outline World Wide Web accessibility requirements (CAST, 2004).

The final document that emerged from the expert review was placed on a one-page review form that can be used during sensitivity reviews. Using this tool, design features such as item respect for the diversity of the assessment population (i.e., bias), readability of text, formatting of text, evaluation of visuals and diagrams, and the feasibility of accommodated formats are all investigated using this form. When placed against the "life of an ELA item" (see Figure 3), this process may potentially be inserted in the boxes entitled "Reading passages brought to item bias committee," "Draft items brought to content committees," "Revised items brought back to content committees," and "Items reviewed by state DOE staff (depending on state)."

Sample: Item Review

To date, the item review process described here has been used twice with states, once as part of an ongoing review of items and once as part of a state-mandated evaluation of an assessment. The first review was largely for informational purposes and future planning. As part of the pilot process of the form, one state assessment entity agreed to let researchers review items using the "Considerations" of item review forms. Findings were reported to the state Technical Advisory Committee and were taken as points to ponder for future iterations of tests, although no action occurred as a result of the process. Feedback from this process indicated that item review was an essential part of the item development process, but that item review alone may be a risky process, as reviewer interpretation can both detect and overidentify bias.

The second use of the protocols occurred as part of a state-mandated evaluation plan of a large-scale assessment. In this case, operational items were reviewed by 24 teachers (teachers were mathematics and language arts content teachers, special education teachers, and staff members at schools for the blind and deaf). Reviews occurred in two parts of the state. Feedback was reported and considered by the state, and the overall evaluation (which included an alignment study, opportunity to learn study, and curriculum studies) won an award for its robust coverage of the assessment.

Procedures and Analysis: Item Review

In the state evaluation review, experts individually completed forms with items then discussed their recommendations. When individual forms were complete, groups shared results and reached consensus on questionable items. All teacher comments were categorized as "minor" or "major" issues. Minor issues were noted as possible issues for vendors to consider. These included visual formatting issues, wording on particular questions, or excessive wordiness on problems. Major issues required approval by all reviewers and were communicated as important and immediate items on which to take action. Major issues included items which had unclear or inappropriate constructs, culturally offensive or irrelevant material, or items for which students have not had an opportunity to learn (according to state standards of curriculum).

Results: Item Review

Results from the state's evaluation review indicated that some mathematics

items had potentially problematic formatting issues (small text, presentation of equations, etc.) and the combination of language arts passages were viewed as overly long, with unclear visuals (visuals did not appear to enhance comprehension and may have been distracting) (Wise et al., 2005). Teachers recommended formatting changes for the mathematics test and a variety of passage lengths for the language arts test (alternating between short, medium, and long passages rather than all long passages).

Following the exercise, an evaluation of the same state's UDA item review process was conducted. Ninety-four percent (94%) of teachers rated the training provided by authors and follow-up item review exercise as "very useful." The remaining 6% said the tool was "useful." By evaluating the UDA item review process as "useful," teachers agreed that the item review process (whereby teachers received training in UDA *Elements* then reviewed items for fidelity to *Elements)* was a useful way to improve assessment practices by giving experts an opportunity to screen items. Teachers also commented that the process was informational because it provided them upfront knowledge about what design issues may be problematic for students. The work of teachers can be couched in two validity arguments. First, by improving the formatting of mathematics items so that formulas are easier to distinguish, tests may provide states with a more accurate estimate of student mathematics ability (this ability may have been obscured by items without clearly distinguishable formulas). Second, the reading ability of students who tire easily (e.g., students with learning disabilities and English language learners) may be more accurately measured by tests that have a wide range of passage lengths. Shortening the length of some passages, according to teachers, would better capture the abilities of struggling readers without changing the construct assessed.

Statistical Analyses

Statistical analysis of item behavior is standard practice in the development of any assessment. Typically, states use descriptive statistics to ensure that tests produce a normal distribution of scores on overall assessments. Similarly, states use techniques like differential item functioning (DIF) analysis to determine if particular items are behaving in a way that would not be expected based on a student's overall score. Such techniques are used to detect bias against particular populations. We argue that statistical analysis, while common practice, is part of the UDA process. Below we highlight approaches that are likely well known by readers of this journal, but are an important part of the UDA process.

Sample: Statistical Analysis

After sensitivity and content reviews, items still face more scrutiny before becoming operational. As seen in the "life of an ELA item" in Figure 1, items that survive the item review process are typically field tested on a sample of students throughout the state. Test developers use field testing to measure the statistical parameters of items before inclusion in live tests (AERA, APA, NCME, 1999). Johnstone, Thompson, Moen, Bolt, and Kato (2005) analyzed data from a large Midwestern state's large-scale mathematics data. These data included student scores, item level data, and information on disability status.

Procedures and Analysis: Statistical Analysis

As stated above, a common method of analysis for field test items analysis is differential item functioning (DIF). These analyses statistically determine whether particular items are more problematic for particular populations (Linn & Harnsisch, 1981). DIF calculations, however, are limited in their usefulness. Researchers need large numbers of students for calculations to be valid, thus creating questionable results for students from low incidence populations (e.g., students with some disabilities, like visual or hearing impairments). For such populations, statisticians must choose whether to disregard questionable data or rely on single tests that are not powerful in detecting differences between populations. Left with such choices, a reasonable approach is to conduct multiple statistical analyses on field test items to see if patterns emerge.

A second method of determining whether items are accessible to a wide variety of students is to conduct statistical analyses of field and operational tests to determine whether items function differently for focal and reference groups after overall differences in scores are taken into account. As part of a broad universal design approach, Johnstone et al. suggested that statistical analyses, even with limitations due to population size, might be useful in creating a portion of a larger picture about questionable items. The authors analyzed a large data set from a Midwestern state's large-scale assessment data using item ranks, item total correlation, differential item function (DIF) with contingency tables, and differential item functioning (DIF) using item response theory (IRT) approaches. In these methods they compared students without disabilities to students from a variety of disability categories.

For item ranks, Johnstone et al. ranked the p-value of items from different groups to determine whether certain items are more challenging (and potentially biased) for particular students. Item ranking assumes that every item has a particular degree of difficulty and is usually expressed with a p-(probability) statistic. From there, specific cut points were used to determine if an item should be flagged. For example, for item ranks, the authors chose five "ranks" as a rule to flag items by first considering the sampling distribution under the null hypothesis.

In this case, the sampling distribution depended on several quantities including the number of items (in this case, the test had 32 items), item difficulties (because ranks are based on item difficulty p-values), and sample sizes (smaller sample sizes lead to larger variability of rank differences). The null hypothesis was that the rank of the item is the same for both groups (or more strictly, item difficulties are the same for both groups) (De Ayala & Kelley, 1997). Any item that had more than five "rank" difference between the general population and a particular focal population was flagged.

Second, Item Total Correlations and simple t-tests were used to determine whether there was a statistically significant difference ($\alpha < .05$) between target and reference groups for a particular item. If a statistically significant difference was found on a particular item, the magnitude of that difference was tested using Cohen's q (1988) index. The magnitude of q that was large (.50 or higher) and confidence intervals that were low (.20 or lower) indicated that an item should be flagged.

In addition, Mantel-Haenszel Chi-Square statistic ($MH\chi^2$) and log odds

Table 1. Items Considered Potentially Problematic by Group and Analysis Method (Grade 4)

Analysis or Group	Potentially Problematic Items*
All Groups, All Analyses (using decision rule)	N
Analysis	
Item Rank	A
Item Total Correlation	none
DIF (Contingency Tables)	M, N, Z, Y
DIF (IRT)	C, N, V, W
Group	
Learning Disability	O,X,Y
Speech/Language Impairment	X
Cognitive Disability	A, K, L, M, O, S, W, Y, Z
Emotional/Behavioral Disorder	O, AA
Other Health Impairment	G, P, A
Hearing Impairment	T
Partial Sight	M, Z, EE
Blind	none
Physical Impairment	C, N, W, Z
Autism	K, M, N, Z, AA
Traumatic Brain Injury	Q, S
Deaf/Blind	AA
Multidisabled	C, N, O, S, Y

*Because items are secure, they have been coded with letters (for the purposes of security, A ≠ item 1, etc.).

Table 2. Items Considered Potentially Problematic by Group and Analysis Method (Grade 8)

Analysis or Group	Potentially Problematic Items*
All Groups, All Analyses (using decision rule)	L, O, R, X
Analysis	
Item Rank	O, R, X, AA
Item Total Correlation	DD
DIF (Contingency Tables)	H, K, L, O, R, X
DIF (IRT)	L, V, Z
Group	
Learning Disability	E, H, K, O, X, AA, EE
Speech/Language Impairment	D, L, Z
Cognitive Disability	A, D, G, J, K, L, N, O, P, Q, X, CC, DD
Emotional/Behavioral Disorder	G, H, L, O, R, X, AA, DD
Other Health Impairment	Q
Hearing Impairment	I, J, K, Q
Partial Sight	L
Blind	S
Physical Impairment	L, R, EE
Autism	G, I, L, O, P, Z, EE
Traumatic Brain Injury	BB
Deaf / Blind	none
Multidisabled	O, R, W, X

*Because items are secure, they have been coded with letters (for the purposes of security, A ≠ item 1, etc.).

ratio (β_{MH}) for target and comparison groups were calculated. Items for which the chi-square statistic was significant, and D (which represents $-2.35\ \beta_{MH}$) was significantly different from 0, were secondarily analyzed using SPD-X calculations. Large positive results for SPD-X confirmed that an item may have been differentially difficult for target groups. There is no significance test for SPD-X calculations, so SPD-X calculations of .05 was selected as the decision point for flagging items. In this case, SPD-X was used to verify Mantel-Haenszel findings.

IRT approaches had clear cut points. Items with significant DIF and a large difference in item parameters (DIF > .05) were secondarily analyzed with SIBTEST software. Any items that had a beta of greater or equal to .05 and a p-value smaller than .05 were flagged. DIF did not mean that an item is more difficult for one group than for another, but implied that a particular item functioned differentially according to student characteristics that are not related to achievement (e.g., disability, gender, and ethnic background; College Board, 2004).

Once all analyses were completed, decision rules were used to determine whether an item would be a candidate for further review. First, if an item was flagged in half of the analyses for a particular population, that item would be a candidate for further review for the particular population (e.g., if an item was flagged across half of the analyses for students with learning disabilities, it would be appropriate to conduct further research on why the item was problematic for this population). Second, if an item was considered a candidate for further review across half of the populations, it would then be considered a candidate for further review for students with disabilities in general. This approach was an attempt to address the problem of small n size. The authors recognize that more flawed analyses do not produce better data, but our intention was to produce a manageable number of items that could then be examined against other data sources.

Results: Statistical Analysis

Using the DIF procedures, item-total correlation, and item ranking, the researchers found that different items

were flagged by population and analysis method (demonstrating the diversity of findings that emerge from using multiple analysis techniques). For each grade level examined (grade 4 and grade 8), at least one item was flagged by both analysis method and disability group. Item A in Grade 4 and Items L, O, R, and X were flagged in grade 8.

In grade 4, students with Developmental/Cognitive Disability (DCD) had 9 items flagged (see Table 1) while grade 8 students with DCD had 13 items flagged (see Table 2). The particular assessment examined only had 31 items, so such large numbers of flagged items are noteworthy. Readers should note that the particular DCD populations who participated in this study did not take their state's alternate assessment, but were placed in its general assessment. The population of students who had the *no* items flagged were students who were Blind (4th Grade) and Deaf/Blind (8th Grade). Such results indicate where bias may exist among disability populations, and demonstrates the heterogeneity of the group labeled "students with disabilities."

Students with learning disabilities comprise the largest population of students within the category "students with disabilities" and may therefore provide the best population from which to draw inferences from statistical item review data. In Grade 4, three items were flagged for this population (items O, X, and Y). These items did not correspond with the one item that was flagged across populations and analysis methods in Grade 4 (item N). Similarly, in Grade 8, items E, H, K, O, X, AA, EE were flagged. Only one of these items (item X) corresponded with the items flagged overall for Grade 8 (items L, O, R, and X). Such results are important in two ways. First, they demonstrate that the only reliable statistics (data from students with learning disabilities) do not correspond with overall statistics at each grade level (nor do they necessarily respond with students with other disabilities). Because of this, caution should be taken with statistical approaches for low incidence populations. At the same time, methods demonstrate that students with disabilities are not a heterogeneous group, and the potential biases experienced by students with learning disabilities may not be the same as students who are blind, have autism, have cognitive disabilities, etc.

Results of this analysis provide supplemental approaches to current practice, but results that should be used with caution and in addition to other analysis approaches. Such practices may provide insights into item tendencies for students with low incidence populations. Such populations may be students with disabilities or ethnic/cultural groups within a state that have a small population (and who may be equally heterogeneous as students with disabilities).

Think Aloud Techniques

As noted above, statistical techniques are a useful procedure in the UDA process of improving assessments for accessibility. Statistics can highlight *which* items may be problematic on a particular assessment, but provide little insight into *why* such items may be problematic. Think aloud techniques are an approach that allows assessment professionals to understand the thinking and problem-solving approaches students use when attempting test items. Insights gained from such approaches provide insights into how the language, presentation, or tasks associated with an item may distract, confuse, or appropriately challenge students.

Sample

In 2003, we selected six mathematics items in Grades 4 and 8 using an early version of the *Considerations of Universally Designed Assessments* item review form. For this study, the *Considerations* form was still under development, so we used Thompson et al.'s (2002) *Elements of universally designed assessments* as guideposts for flagging issues related to construct-irrelevant variance. We then conducted think aloud methods with 38 Grade 4 students (including 9 students with learning disabilities, 10 students who were deaf or hard of hearing, 6 students with cognitive disabilities, 7 students who were English language learners, and 10 students who were English proficient and did not have disabilities). In addition, we conducted methods with 43 Grade 8 students (including 10 students with learning disabilities, 11 students who were deaf or hard of hearing, 3 students with cognitive disabilities, 9 English language learners and 10 students who were English proficient and did not have disabilities).

Procedures and Analysis

As an item undergoes review and analysis, it is also important to know why an item may be particularly challenging for students. To answer questions about why particular items may be problematic for students, researchers can study student responses to particular items captured while they are taking tests. By using "think aloud" methods, the primary data source is student utterances. Students produce utterances by verbalizing what they are thinking while they are solving problems.

As seen in Figure 3, there are two points at which think aloud methods are appropriate. The first logical point for think aloud methods is when, according to Marion (2006), of "Surviving items brought back to item developers ... some will need significant revision, while others will be eliminated." States and vendors may wish to discriminate between items that have potential design problems (that may cause construct-irrelevant variance) and items that simply have challenging, standards-driven content (which is not a reason to eliminate an item). Conducting think aloud methods are useful for these purposes. Think aloud methods may also be useful prior to including items on field tests as an additional quality control approach. Finally, think aloud methods may be used after field or operational test results are analyzed to determine why particular items may have functioned in a particular way.

This methodology is unique because it seeks to tap the cognitive processes involved with test taking (and the issues related to items that may cause students to have problems) by requesting students to verbalize as they think, or "think aloud." Such methods help determine whether design issues are present in items (Johnstone, Bottsford Miller, & Thompson, 2006) and to determine the validity of testing accommodations (National Research Council, 2004).

Leighton (2004) noted that think aloud methods can be effective for understanding item qualities for assessments, but recognized several limitations to the method, such as the decreased ability of researchers to gain meaningful data from items that are too simple or too challenging for students.

Therefore, soliciting additional information through post-think aloud interviews may be a useful research tool. Branch (2000) and Fontein, Knipers, and Grobe (1993) found that asking post-think aloud interview questions of subjects who struggled with the think aloud process provided valuable information that made think aloud data easier to understand and interpret.

Young students, students with disabilities, and students who are not English proficient may have difficulty with think aloud methods. Therefore, we used a two-step process for think aloud techniques. During the first step, we employed think aloud techniques and probed subjects as infrequently as possible because subjects are easily distracted during problem solving activities. If subjects did not respond for an extended period, we merely encouraged the subject to "keep talking." Neutral cues such as "keep talking" diminish the possibility of leading subjects toward particular responses (Ericsson & Simon, 1995). All field researchers in the study followed the same protocols (as evidenced by videotaped sessions).

Once the think aloud process was complete, we asked follow-up questions. Answers to these questions supplemented any unclear data derived from think aloud protocols. In the study described in Johnstone et al. (2006), there was no set script for follow-up questions. Rather, we took careful field notes during verbalizations and asked students to clarify statements that were unintelligible or needed further elaboration.

For analysis purposes, think aloud methods were merged with universal design literature. For example, when a student verbalized "I don't get why this picture is here," inferences were made about the appropriateness of graphics. Also, when a student looked at a problem and said "I don't get what I'm supposed to do here" or went a completely different direction than the tests' authors had intended, inferences were made about the intuitiveness of directions or clarity of construct. Videotaped protocols were coded using a standard coding sheet. Each video was reviewed by at least two reviewers, and every tenth tape was reviewed by four reviewers. If interrater agreement was not reached on any portion of any coding sheet, raters discussed results until consensus was reached (e.g., if there was a disagreement on which UD *Element*, if any, that a student response revealed, all reviewers would rereview the tape until agreement was reached). The coding sheet included prompts for researchers about student responses, such as accuracy of reading, problem solving, and calculation. Results were then analyzed for strengths or shortcomings using Thompson et al.'s (2002) *Elements of universally designed assessments* as a template. Specifically, researchers attempted to glean information from protocols on: precisely defined constructs, bias-free items, clear and intuitive instructions, comprehensible language, and legible print and graphics.

Results: Think Aloud Methods

Johnstone et al. (2006) reviewed six mathematics items each in Grades 4 and 8. These items are secure, so it is not possible to report results based on the actual items. Rather, results are reported by *Element* below. Readers should note that think aloud methods were not effective in determining why students struggled on two items (Grade 4, Item 4 and Grade 8, Item 5) and were effective at determining that challenging content, not design features, was the cause of error for Grade 4, Item 1. All other items appeared to have some design feature that led to misunderstanding for some students, as reported below.

Precisely Defined Constructs

Think aloud methods detected design issues for three items: Grade 4, items 3 and 6, and Grade 8, item 3. According to the data, each of the items was challenging to students because of an unclear construct. Students who answered Grade 4, item 3 incorrectly failed to understand a message that indicated that the item required a two-part solution. Students then answered the item as if it was a one-step item. Similarly, Grade 4 item 6 required students to use a map. The item contained construct-irrelevant variance (Haladyna & Downing, 2004) that was identified by student utterances, including confusion about which numbers to add and confusion about how the map worked (arrows were used in two different ways on the surface of the map). Finally, analysis of Grade 8, item 3 revealed that eight students who provided incorrect answers were confused or distracted by some aspect of the item.

Accessible, Nonbiased Items

Think aloud methods detected issues related to bias in Grade 8, items 4 and 6. Sixteen of 43 participants answered Grade 8, item 3 incorrectly. For four of these students, failure to respond correctly to this item appeared to be due to inexperience with this particular type of item (the item required students to calculate a missing property from a chart). Results from Grade 8, item 6 revealed that students were unfamiliar with the context from which a mathematics item was based. Because of this, students did not recognize the relation of examples described in the item (students thought items were part of a larger entity rather than separate items that needed calculation).

Simple, Clear, and Intuitive Instructions

Four of the 12 items selected for this research appeared to have issues related to simple, clear, or intuitive instructions (Grade 4, items 3, and 5; Grade 8, items 2 and 3). Ten out of 39 students in the sample answered Grade 4, item 3 incorrectly. The content of the instructions confused five of these students, who solved the problem by placing numbers in an ascending order, rather than the descending order that the item required.

Only 7 out of 40 students answered Grade 4, item 5 correctly. In this item, instructions were given at the top and bottom of the page, with a workspace in the middle. Four students stopped after reading the directions on top of the page and asked researchers what they were supposed to do (the item did not provide such direction). Twenty out of 43 students in the sample answered Grade 8, item 1 correctly. Ten students were unclear of the intention of italicizing a particular word. Finally, during protocols for Grade 8, item 3, four students misinterpreted the meaning of what the item was asking.

Maximum Readability and Comprehensibility

Think aloud methods detected issues related to readability or comprehensibility for three items (Grade 4, item 3; Grade 8, items 1 and 2). Two of the 10 students who were not successful on Grade 4, item 3 had difficulties related to the text in the item. These students were unfamiliar with the keyword in the

Table 3. Fourth Grade Item Results and Conclusions

Item	Results	Conclusions
4th Grade, Item 1	Content too difficult for students	
4th Grade, Item 2	Students were confused or did not understand how the word "more" associated with correct answer.	Detected *Element 5 (Simple, Clear and Intuitive Instructions and Procedures)* deficiencies, but demonstrated that *Element 7 (Maximum Legibility)* may not be enough to facilitate student understanding.
4th Grade, Item 3	Students appeared unclear on words "most" and "fewest."	Detected deficiencies in *Element 2 (Precisely Defined Constructs)*, *Element 5 (Simple, Clear and Intuitive Instructions and Procedures)*, and *Element 6 (Maximum Readability and Comprehensibility)*.
4th Grade, Item 4	Data inconclusive as to why students had difficulties.	Think aloud data cannot detect design deficiencies for all items.
4th Grade, Item 5	Instructions confusing to some students. Think aloud data may have confounded confusion.	Detected deficiencies in *Element 5 (Simple, Clear and Intuitive Instructions and Procedures)*. Think aloud methods should be carefully approached in items that ask students to "explain."
4th Grade, Item 6	Students misread map.	Detected deficiencies in *Element 2 (Precisely Defined Constructs)* and *Element 7 (Maximum Legibility)*.

Source: Johnstone et al. (2006).

Table 4. Eighth Grade Item Results and Conclusions

Item	Results	Conclusions
8th Grade, Item 1	Language in instructions confused students.	Detected deficiencies in *Element 5 (Simple, Clear and Intuitive Instructions and Procedures)* and *Element 6 (Maximum Readability and Comprehensibility)*.
8th Grade, Item 2	Students confused by instructions asking them to pick "best" choice.	Detected deficiencies in *Element 5 (Simple, Clear and Intuitive Instructions and Procedures)* and *Element 6 (Maximum Readability and Comprehensibility)*.
8th Grade, Item 3	Construct unclear to some students.	Detected deficiencies in *Element 2 (Precisely Defined Constructs)* and *Element 5 (Simple, Clear, and Intuitive Instructions)*.
8th Grade, Item 4	Students unable to answer item correctly because they were unfamiliar with content.	Think aloud method was effective in demonstrating opportunity to learn (OTL) discrepancies between students. Detected *Element 3 (Accessible, Non-biased Items)* deficiencies.
8th Grade, Item 5	Data inconclusive as to why students had difficulties.	Think aloud data cannot detect design deficiencies for all items.
8th Grade, Item 6	Language used in item confusing or unfamiliar to students.	Detected deficiencies in *Element 3 (Accessible, Non-biased Items)*, *Element 6(Maximum Readability and Comprehensibility)*, and *Element 7 (Maximum Legibility)*.

Source: Johnstone et al. (2006).

item. One student crossed out the keyword and replaced it with a word she thought to be a synonym. Only 18 out of 42 students answered Grade 8, item 1 correctly. Students had difficulty with reading both the construct and nonconstruct vocabulary fluently. Grade 8, Item 2 was also challenging to one student because one of the terms used in the item had a double meaning (as both a mathematical and nonmathematical term).

Maximum Legibility

Finally, think aloud methods detected legibility issues in two items (Grade 4, item 6; Grade 8, item 6). Five students who answered Grade 4, item 6 incorrectly did so because of the font selected on the page (the numerals "1" and "7" looked similar and confused students). Grade 8, item 6 had similar editorial issues. In this item, fractions were presented in two different ways.

Overall, think aloud methods, when analyzed using Thompson et al.'s (2002) universal design template, were very helpful in determining why students made errors on items. By using think aloud studies, researchers could pinpoint the source of error in an item and determine if the error was related to construct or nonconstruct stimuli (see Tables 3 and 4 specific item results). Error analyses of student incorrect responses demonstrated

that a higher percentage of students with learning disabilities (14%), students who were deaf (12%), and English language learners (17%) were distracted more often by construct-irrelevant text and images than their nondisabled, English proficient peers (8%).

Students from all populations except students with cognitive disabilities were able to meaningfully participate in think aloud activities. Future studies with this population would need to include a backup plan for students who were unable to verbalize in a meaningful way during think aloud protocols. For these students, a familiar facilitator (such as a teacher) may improve think aloud verbalizations. Alternatively, Van Someren et al. (1994) suggested prompting students to act like teachers and "teach" facilitators how to complete items (thus verbally demonstrating areas of focus in the item).

Discussion: Implications for Practitioners

In this article, we proposed that three specific approaches present an opportunity for assessment personnel to efficiently, yet dependably determine items that may be problematic to particular populations. By doing this, item developers can pause at different stages of an item's development to ensure that items conform to generally accepted notions of "universal design."

The goal of this article was to propose analysis methods at various stages of the "life of an item" (Marion, 2006). By framing item reviews through the lens of universal design, items can be reviewed for accessibility and comprehensibility for all students. A potential next step to this article is to use all item review methods (expert review, quantitative analysis, and think aloud techniques) collectively in a state's item development processes. By doing this, states and vendors could better place claim that their assessments are universally designed. In addition, the three methods proposed, when used together, will triangulate decisions made on items throughout the item review process. Items that are deemed questionable can be analyzed statistically (if they reach field testing) and can undergo think aloud methods at any time in the development process. Further experimental research, such as determining whether items better discriminate if subjected (and changed) by item review processes would also be valuable.

The universal design approach is still under development and will need more research to determine its effectiveness. In particular, more research is needed to determine how different pieces of evidence should be weighed. It is obvious that if an item appears questionable in three analysis methods, it should be flagged for possible removal. If items appear questionable in only one or two methods, however, evidence is less concrete. In cases such as these, other factors might be considered, including: the level of alignment an item has with standards, possible accommodations that could ameliorate problems caused by the item, an examination of the access skills needed to solve the item, or alternate items that might be used to test the same construct.

This article, then, should be considered a first step in combining information from three item analysis techniques based on universal design approaches. Each technique has obvious limitations (item review is prone to human error, statistical methods are questionable with low incidence populations, and think aloud techniques are difficult to generalize because of small sample sizes and are expensive). Employing a multistage and multimethod approach to item review, however, may provide states with a more trustworthy, cost-effective, and standardized way to review items. Items undergo tremendous scrutiny from the time they are written to the time they are operational. This article proposed methods that can be used with existing practices, to inform item reviewers, states, and vendors about universal design of assessment at the same time.

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